

Advanced (Carolina Reaper)

Q1. Simplify $\sqrt{64} + \sqrt{16}$

- (A) $8 + 4$
- (B) $4 + 8$
- (C) 8
- (D) 12
- (E) 16

Q2. Simplify $\sqrt{25} \cdot \sqrt{9}$

- (A) $3 \cdot 5$
- (B) $5 \cdot 3$
- (C) 15
- (D) 45
- (E) 30

Q3. A furniture maker needs to cut a $12\sqrt{2}$ ft long board into two equal pieces. How long is each piece?

- (A) $6\sqrt{2}$ ft
- (B) 6 ft
- (C) 8 ft
- (D) 12 ft
- (E) $8\sqrt{2}$ ft

Q4. Simplify $\frac{\sqrt{36}}{\sqrt{4}}$

- (A) $\frac{6}{2}$
- (B) $\frac{3}{2}$
- (C) $\frac{9}{2}$
- (D) 3
- (E) 6

Q5. Simplify $\sqrt{16} - \sqrt{4}$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) $2 - 2$

Q6. Simplify $(\sqrt{9} + \sqrt{16})^2$

- (A) $(3 + 4)^2$
- (B) 7^2
- (C) 49
- (D) 25
- (E) 16

Q7. Simplify $\frac{\sqrt{8}}{\sqrt{2}}$

- (A) $\sqrt{4}$
- (B) $\sqrt{2}$
- (C) 2
- (D) 4
- (E) 8

Q8. A clockmaker has $6\sqrt{3}$ hours to complete a project. If she works $2\sqrt{3}$ hours per day, how many days will she take to complete the project?

- (A) 2 days
- (B) 3 days
- (C) 6 days
- (D) $2\sqrt{3}$ days
- (E) $3\sqrt{3}$ days

Open Ended Questions (FRQ)

Q9. A builder is constructing a roof with a $10\sqrt{2}$ ft long rafter. If he needs to cut it into three equal pieces, show all steps to find the length of each piece.

Q10. A water tank can hold $24\sqrt{3}$ gallons of water. If $8\sqrt{3}$ gallons of water are already in the tank, what fraction of the tank is filled? Simplify your answer.

Chef's Tips

- Read each question carefully before answering
- Use the given space for Free Response Questions
- Check your work before submitting